

RONNEBY, Sweden

WMO: 026640

Lat: 56.267N

Lon: 15.268E

Elev: 58

StdP: 100.63

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-9.7	-15.0	1.0	-11.2	-12.2	1.3	-8.0	10.9	3.8	9.7	3.0	2.2	10	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	26.6	18.8	24.9	18.1	23.1	17.4	20.4	24.2	19.3	22.9	18.3	21.8	4.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.9	22.1	17.9	13.0	21.1	16.9	12.2	20.0	58.7	24.3	55.0	22.7	51.9	21.8	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	7.0	-16.7	29.3	3.8	2.0	-19.5	30.8	-21.7	31.9	-23.8	33.0	-26.5	34.5		
(5)			-17.0	22.0	3.8	1.2	-19.7	22.8	-22.0	23.5	-24.1	24.2	-26.8	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-0.5	-0.2	1.9	6.1	10.6	14.3	16.9	16.5	12.7	8.0	4.0	0.8		
	DBStd	7.16	4.24	4.07	3.74	3.43	3.49	2.84	2.73	2.63	2.81	3.40	3.63	4.54		
	HDD10.0	1577	325	287	253	124	34	2	0	0	9	78	181	284		
	HDD18.3	3937	583	520	511	366	241	126	61	68	170	319	431	542		
	CDD10.0	716	0	0	0	8	52	131	216	202	89	17	1	0		
	CDD18.3	36	0	0	0	0	1	4	18	12	1	0	0	0		
	CDH23.3	327	0	0	0	1	11	47	180	87	2	0	0	0		
	CDH26.7	44	0	0	0	0	1	6	29	9	0	0	0	0		
(14) Wind	WSAvg	3.3	3.5	3.5	3.5	3.4	3.4	3.3	3.0	2.9	3.1	3.2	3.2	3.3		
(15) Precipitation	PrecAvg	670	50	42	31	37	53	62	66	67	61	67	72	60		
	PrecMax	839	84	83	71	76	132	159	139	153	183	150	151	159		
	PrecMin	487	0	15	4	3	9	0	4	22	12	4	12	25		
	PrecStd	81	24	22	17	18	25	31	31	36	45	35	32	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	9.2	14.3	21.1	24.7	27.1	29.3	27.8	23.0	16.9	12.6	10.1		
		MCWB	7.6	7.1	8.7	13.6	17.5	18.4	19.6	19.7	17.9	14.2	11.4	9.4		
	2%	DB	7.0	7.8	11.8	17.4	22.2	24.6	27.2	25.8	20.9	15.1	11.1	8.8		
		MCWB	6.3	6.5	7.5	11.1	15.3	17.4	19.2	18.4	16.5	13.4	10.2	8.0		
	5%	DB	5.8	6.4	9.1	15.0	20.0	22.5	25.4	23.9	19.2	14.0	9.9	7.7		
		MCWB	5.2	5.3	6.0	9.9	13.9	16.3	18.5	18.3	15.5	12.7	9.2	7.1		
	10%	DB	4.8	5.1	7.2	12.9	17.9	20.8	23.6	22.2	17.7	13.0	8.9	6.6		
		MCWB	4.3	4.2	5.1	8.5	12.6	15.3	17.7	17.7	14.7	11.8	8.4	6.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	8.0	10.0	14.3	18.4	20.4	21.9	21.6	19.0	15.1	11.8	9.6		
		MCDB	8.3	8.4	13.0	19.9	23.6	25.3	26.6	24.7	21.7	16.2	12.3	9.9		
	2%	WB	6.4	6.7	8.0	12.0	16.3	18.5	20.6	20.2	17.2	13.9	10.4	8.1		
		MCDB	6.7	7.4	10.7	16.4	20.4	22.9	25.1	23.2	19.6	14.7	11.0	8.5		
	5%	WB	5.2	5.4	6.5	10.5	14.8	17.1	19.5	19.1	16.2	13.0	9.2	7.1		
		MCDB	5.6	6.1	8.5	14.1	18.7	21.0	23.6	22.0	18.2	13.7	9.6	7.5		
	10%	WB	4.2	4.3	5.3	9.1	13.3	15.9	18.4	18.2	15.2	12.1	8.4	6.1		
		MCDB	4.6	5.1	6.9	12.3	16.8	19.6	22.3	21.2	17.0	12.9	8.8	6.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.5	7.8	10.2	10.7	10.2	10.0	9.6	8.7	6.8	5.0	4.7		
		MCDBR	4.4	6.2	11.3	14.0	14.0	13.3	13.6	12.0	10.6	7.4	4.8	4.6		
	5% WB	MCWBR	4.1	5.0	7.6	8.7	8.1	7.4	6.7	6.4	6.7	5.7	4.5	4.5		
		MCWBR	4.0	5.5	9.9	12.6	12.0	11.8	11.4	10.2	9.3	6.7	4.6	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.333	0.351	0.384	0.382	0.374	0.398	0.397	0.367	0.355	0.335	0.298		
		taud	2.289	2.319	2.343	2.320	2.383	2.434	2.385	2.388	2.452	2.402	2.327	2.256		
	Ebn at Noon		590	723	809	833	858	870	838	812	788	691	547	494		
		Edh at Noon	53	75	93	110	110	106	109	102	84	69	51	42		
(44) All-Sky Solar Radiation	RadAvg		0.41	1.02	2.45	4.09	5.37	5.70	5.21	4.21	2.93	1.39	0.53	0.29		
	RadStd		0.07	0.18	0.24	0.50	0.54	0.38	0.62	0.37	0.34	0.24	0.10	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	+1.88	N/A	N/A	N/A	N/A	-142	N/A	N/A
(47)	Regional (46 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-162	N/A	N/A

Nomenclature: See separate page